ACME METALS

Digital Material Passport

ID **DMP-2025-8842-1-002** - *Version* **1.0.0**
Issue Date **2025-03-18** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
Nordstahl Schmiedewerke GmbH Werkstraße 12 44135 Dortmund DE info@nordstahl-schmiede.example.com	**ACME Steel Trading GmbH** Handelsweg 5c 76199 Karlsruhe DE customer@example.com	**Order** PO-2025-8842 · **19000 kg** **Delivery** DN-2025-8842 · **3800 kg**

Product

Product Name	Forged round bars, rough turned
Batch ID	BATCH-2025-8842
Surface Condition	Rough turned
Delivery Condition	+FP - (Ferrite/pearlite annealed)
Weight	9.5 t
Diameter Tolerance	+3/-0 mm
Specification	8842-1/ GENERIC - FRT - Rev 1.0 - 2025-03-18
Name (EN 10027-1)	18CrNiMo7-6 +FP
Number (EN 10027-2)	1.6587
Shape	RoundBar - D 500 mm - Length 1500 mm
Marking	Head of bar stamped with grade, size, heat No., weight, forging No. Marking has to allow clear identification
Packaging	Bulk, max. weight of bar 13,5 mt
Coloring	Yellow/black
Product Norm	ISO 683-3
Product Norm	ISO 6336-5

Heat Treatment

Process: **Annealing** - *Lot:* HT-LOT-2025-8842 - *Furnace:* F-12 - *Date:* 2025-03-10

Stage	*Temperature*	*Duration*	*Cooling*	*Atmosphere*
Soaking	650 C	6 h		Air

Chemical Analysis

Heat Number **HT-2025-4521** - *Melting Process* **EAF+LF+VD** - *Casting Method* **ContinuousCasting** - *Sample Location* **Ladle**

Symbol	C	Si	Mn	P	S	Cr	Ni	Mo	Cu	Ti	Al	V	Sn	N	Ca	O	H	F1	F2
Unit	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	ppm	%	-
Min	0.15	0.15	0.5			1.5	1.4	0.25			0.02			0.01					2
Max	0.21	0.4	0.9	0.02	0.008	1.8	1.7	0.35	0.25	0.005	0.05		0.03	0.017	0.0025	25	2	0.5	3.5
Actual	0.18	0.27	0.65	0.01	0.004	1.65	1.55	0.3	0.12	0.003	0.03	0.005	0.012	0.013	0.0015	12	1.2	0.24	2.31

F1 = Cu+10*Sn: 0.24% **F2** = Al/N: 2.31-

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*
Tensile Strength[1] Specimen: 1/4T, L	Rm	1265 N/mm²	1200 N/mm²		
Yield Strength[1] Specimen: 1/4T, L	Rp0,2	905 N/mm²	850 N/mm²		
Elongation[1] Specimen: 1/4T, L	A5	13.5 %			
Reduction of area[1] Specimen: 1/4T, L	Z	52 %			

Impact test ISO-V
+20°C on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling
Specimen: 1/4T, L

No.	1	2	3	*Average*	*Min*	*Spec Min*
Value [J]	58	62	60	60	58	40

Impact test ISO-V
-30°C on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling
Specimen: 1/4T, L

No.	1	2	3	*Average*	*Min*
Value [J]	45	48	47	46.7	45

	Actual	Minimum	Maximum
Surface hardness	185 HB	159 HB	207 HB
Core hardness	178 HB	159 HB	207 HB

[1] on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling

Physical Properties

Property	*Actual*	*Target/Min*	*Maximum*	*Method*

Grain size	7	6		ISO 643
Purity Plate II, inspected area approx. 200 mm²				ISO 4967 method B

Inclusion type	A fine	A thick	B fine	B thick	C fine	C thick	D fine	D thick	DS
Value	2	1.5	1.5	1	1	0.5	1.5	1	1
Max	3	3	2.5	1.5	2.5	1.5	2	1.5	2

Purity K4	14		22	DIN 50602
Straightness	1.5 mm/m		2.5 mm/m	
Reduction ratio	6	4		
Residual magnetism	8 A/cm		15 A/cm	
Structure	Ferrite/pearlite			

✓ Fully formed structure ✓ Protected from reoxidation

Supplementary Tests

Test	*Result / limits*	*Method*
Ultrasonic test 100%	Yes - Pass	EN 10228-3 Type 1a class 3
Surface test 100% visually checked, clean surface free from defects	Yes - 100% visually checked, clean surface free from defects	
Radioactivity	Yes - max 100 Bq/kg	
Square cut obliquity max. 2% of diameter, no centerholes	Yes - Max. 2% of diameter, no centerholes	

✓ Free from mercury ✓ Free from cadmium ✓ 100% checked ✓ Sawn, free from burr

Hardenability – Jominy (ISO 642), values in HRC - +HH band; tested per ISO 642 or calculated values

Distance - from - quenched - end (mm)	1.5	3	5	7	9	11	13	15	20	25	30	35	40	45	50
Value [HRC]	46	46	45	44	44	44	42	42	40	39	38	37	37	30	30
Min	43	43	42	41	40	40	39	38	36	35	34	33	33		
Max	48	48	48	48	47	47	46	46	44	43	42	41	41		

Validation

We confirm that the goods described above comply with the requirements of the order and this inspection document was prepared in accordance with EN 10204:2004, type 3.1.

✓ 100% of the material is from Chinese sources

✓ Material is of non-Russian origin *(EU Regulation No. 833/2014)*

Validated by **Dr. Lena Krause**, Head of Quality Assurance, Quality Assurance - 2025-03-18